

LESSON 3-7 • SESSION 2 • PRACTICE • FORM B

Multi-Step System Conversions and Dosage

Learning target – I can solve a multi-step problem that requires converting units before using a rate. · 6.RP.A.3d

REMEMBER

- Start with the amount you are given. Change it to the unit the **next step** needs. Then use the **rate**.
- Write the **unit** after every answer: lb, then kg, then mg, then mL.
- For area, multiply length $\times$ width, then divide by what one bag or can covers. Round **up** for whole items.

WORKED EXAMPLE

A cat weighs 11 lb. The dose is 3 mg per kg. The syrup has 5 mg per mL. (1 kg $\approx$ 2.2 lb)

Step 1 $11 \div 2.2 = 5$ kg.

Step 2 $5 \text{ kg} \times 3 \text{ mg per kg} = 15 \text{ mg}$.

Step 3 $15 \text{ mg} \div 5 \text{ mg per mL} = 3 \text{ mL}$.

Answer 3 mL of syrup

WORDS multi-step (*de varios pasos*) · dose (*dosis*) · per (*por*)

PRACTICE*show your work · label units*

1. **Luna the Cat.** Complete each step. (1 kg $\approx$ 2.2 lb)

a. $22 \text{ lb} \approx$ _____ kg

b. $10 \text{ kg} \times 6 \text{ mg per kg} =$ _____ mg

c. $60 \text{ mg} \div 12 \text{ mg per mL} =$ _____ mL

d. $30 \text{ kg} \approx$ _____ lb

2. **Pony Medicine.** A pony weighs 176 lb. The medicine dose is 3 mg per kg of body weight. (1 kg $\approx$ 2.2 lb)

Hint Change pounds to kilograms before you use the dose.

a. Weight in kilograms: _____ kg

b. Total dose: _____ mg

Show your work.

3. **Elephant Calf.** A baby elephant weighs 120 kg. (1 kg $\approx$ 2.2 lb, 1 ton = 2,000 lb)

Hint Multiply by 2.2, then subtract from 2,000.

a. Weight in pounds: _____ lb

b. Pounds more to reach 1 ton: _____ lb

Show your work.

4. **Gas Tank.** A car's tank holds 76 L. Gas costs \$3.80 per gallon. (1 gal $\approx$ 3.8 L)

a. Tank in gallons: _____ gal

b. Cost to fill: _____

Show your work.

5. **Butter.** A recipe calls for 196 grams of butter. About how many ounces is that? (1 oz $\approx$ 28 g)

Hint An ounce is heavier than a gram, so divide.

Show your work.

Answer _____ oz

6. **Dog Medicine.** A dog weighs 88 lb. The dose is 5 mg per kg. Find the dose. (1 kg $\approx$ 2.2 lb)

Hint Kilograms first, then multiply by 5 mg.

Show your work.

Answer _____ mg

7. **Travel Cost.** A trip is 300 km. A car uses 1 L of fuel every 12 km. Fuel costs \$1.60 per liter.

Hint Divide the distance by 12 km per liter.

a. Liters used: _____ L

b. Fuel cost: _____

Show your work.

8. **Class Garden.** A garden bed is 8 ft by 9 ft. One bag of soil covers 8 square feet and costs \$3.75.

a. Area of the bed: _____ ft²

b. Bags needed: _____ bags

c. Total cost: _____

d. What changes if one bag covers 12 square feet? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Priya's Work.** Priya converted a 55-lb goat to kilograms by multiplying: $55 \times 2.2 = 121$ kg. Explain why 121 kg is impossible. Then find the correct weight. (1 kg $\approx$ 2.2 lb)

TEST PRACTICE

choose the best answer

10. A 66-pound dog is about how many kilograms? (1 kg $\approx$ 2.2 lb)

A 145.2

B 30

C 68.2

D 63.8

11. At 4 mg per kg, a 15-kg animal needs –

A 60 mg

B 19 mg

C 3.75 mg

D 11 mg

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

440 Z	10 W	200 T	1,736 I	\$76.00 G	240 E	528 Q	\$33.75 S	7 H	\$288.80 X
#1a	#2b	#3b	#4b	#5	#6	#8c			

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____